

Mapping of research questions, objectives and evaluation evidence

Primary RQ: How can an integrated LLM-driven framework improve the effectiveness, efficiency and reliability of continuous testing in CI/CD pipelines?

RQ1: How can LLMs improve continuous testing in CI/CD pipelines?

RQ2: How effective is LLM-based predictive test selection compared with traditional methods?

RO1: Design modular agents for test generation, predictive selection and self-healing.

RO2: Empirically compare coverage, defect detection and execution efficiency.

Evidence: prototype behaviour + controlled traditional-versus-agentic experiment